AI Assisted Coding

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Batch : 34

Course Code : 23CS002PC304

Assignment Number : 17.2

Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

Task Description – 1(Student Academic Data Cleaning)

• Task: Use AI assistance to generate a Python script that cleans and prepares a student academic dataset for analysis.

Instructions:

• Handle missing values in marks, attendance, and department.

• Remove duplicate student records based on student ID.

• Standardize text fields such as department names.

• Encode categorical variables like gender and department.

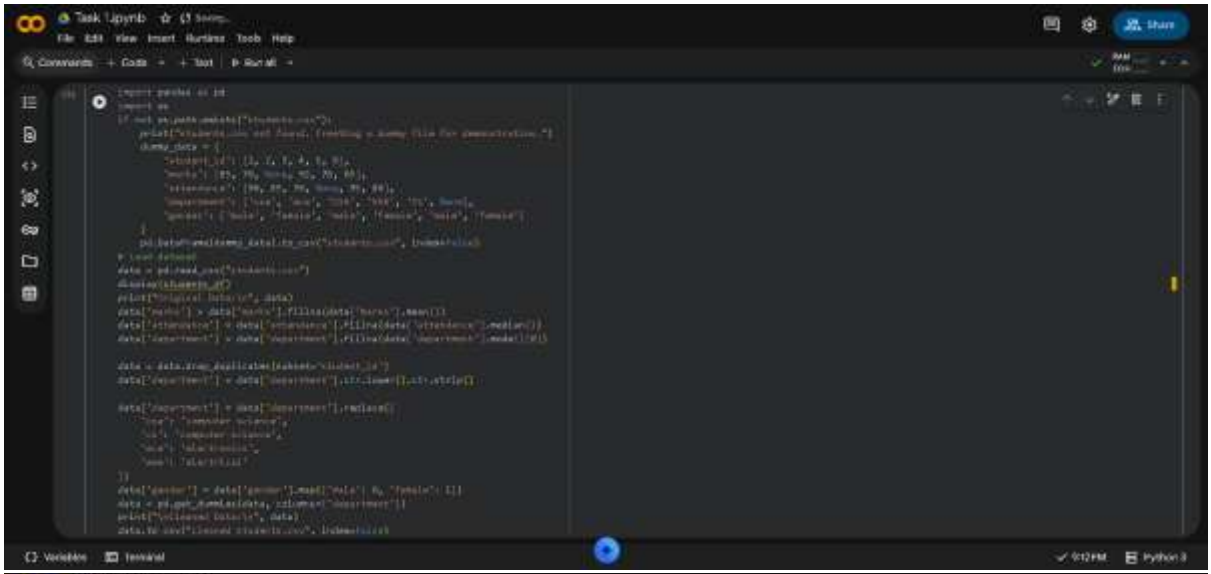
Sample Code:

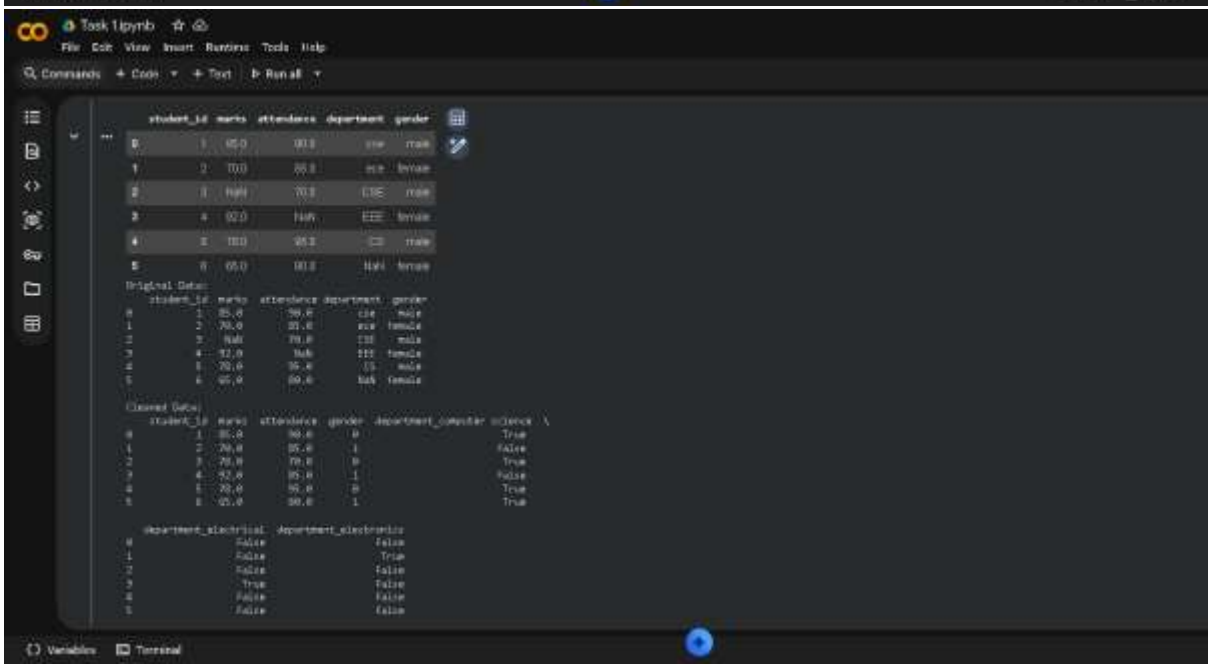
```
import pandas as pd

data = pd.read_csv("students.csv")

print(data.head())
```

Expected Output:

• A cleaned and encoded Pandas DataFrame suitable for academic

performance analysis.





Task Description – 2 (Financial Transaction Data Preprocessing)

• **Task:** Preprocess financial transaction data using AI-generated Python code.

Instructions:

• Convert transaction date columns to datetime format.

• Create derived features such as transaction month and year.

• Normalize the transaction amount column.

• Identify and handle extreme transaction values.

Sample Code:

import pandas as pd

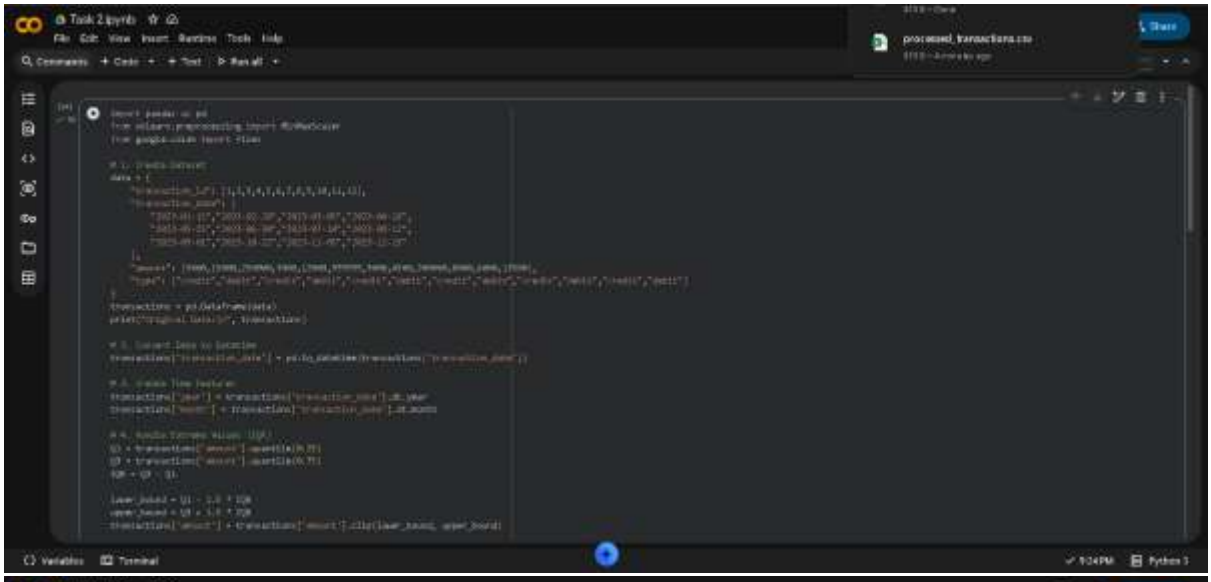
from sklearn.preprocessing import MinMaxScaler

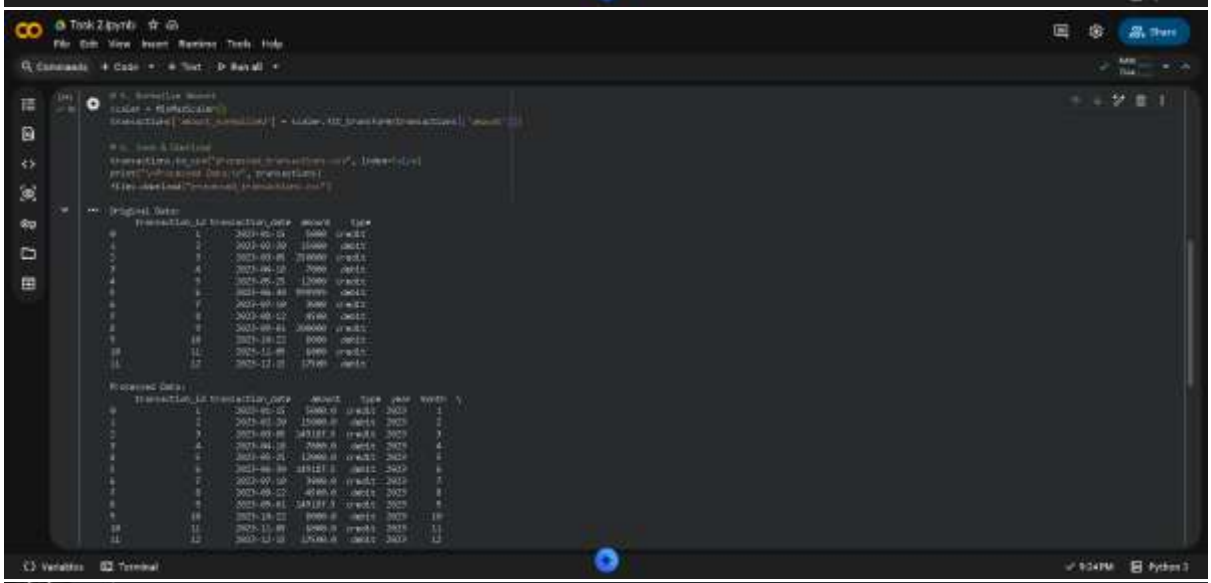
transactions = pd.read_csv("transactions.csv")

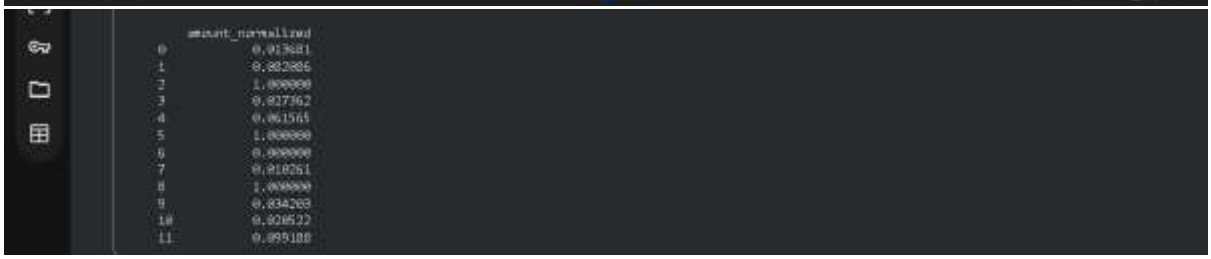
print(transactions.info())

Expected Output:

• A transformed dataset with normalized values and newly generated time based features.







Task 3 – (Environmental Sensor Data Preparation)

• Task: Clean and preprocess environmental sensor data using AI-assisted scripts.

Instructions:

• Handle missing values in temperature, humidity, and air_quality_index.

• Scale numerical features using standard scaling.

• Encode categorical fields such as sensor_status.

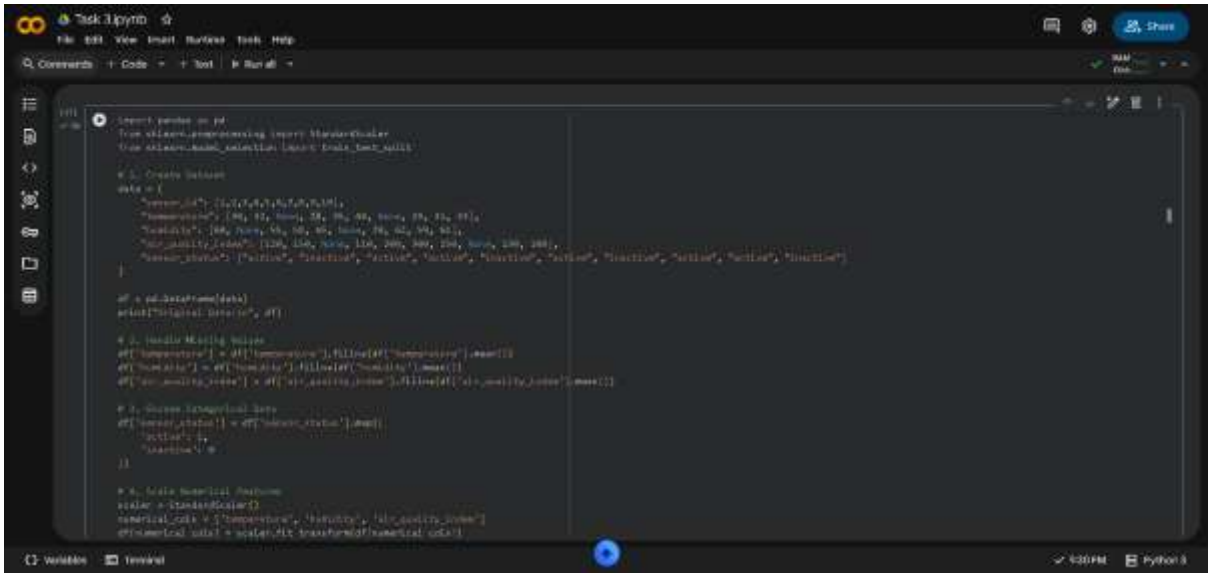
• Split the dataset into training and testing subsets.

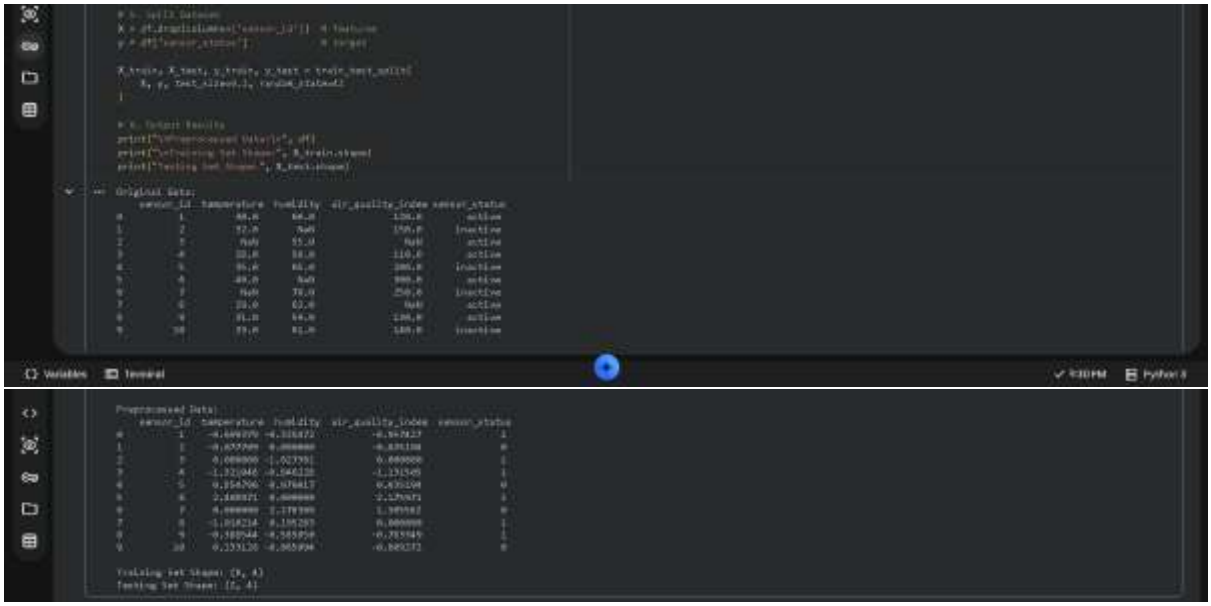
Sample Code:

```
import pandas as pd

from sklearn.model_selection import train_test_split

data = pd.read_csv("sensor_data.csv")
```

Expected Output:

• A fully preprocessed dataset ready for predictive modeling tasks.





Task 4 – (Real-Time Application: Smart City Traffic Data Cleaning)

• Scenario:

A smart city system collects traffic data from multiple sensors, resulting in noisy and inconsistent records.

Requirements:

• Standardize road and location names.

• Fill missing traffic density values using appropriate statistical methods.

• Remove duplicate sensor readings.

• Normalize speed and vehicle count metrics.

• Generate a brief summary comparing dataset quality before and after cleaning

Sample Code:

import pandas as pd

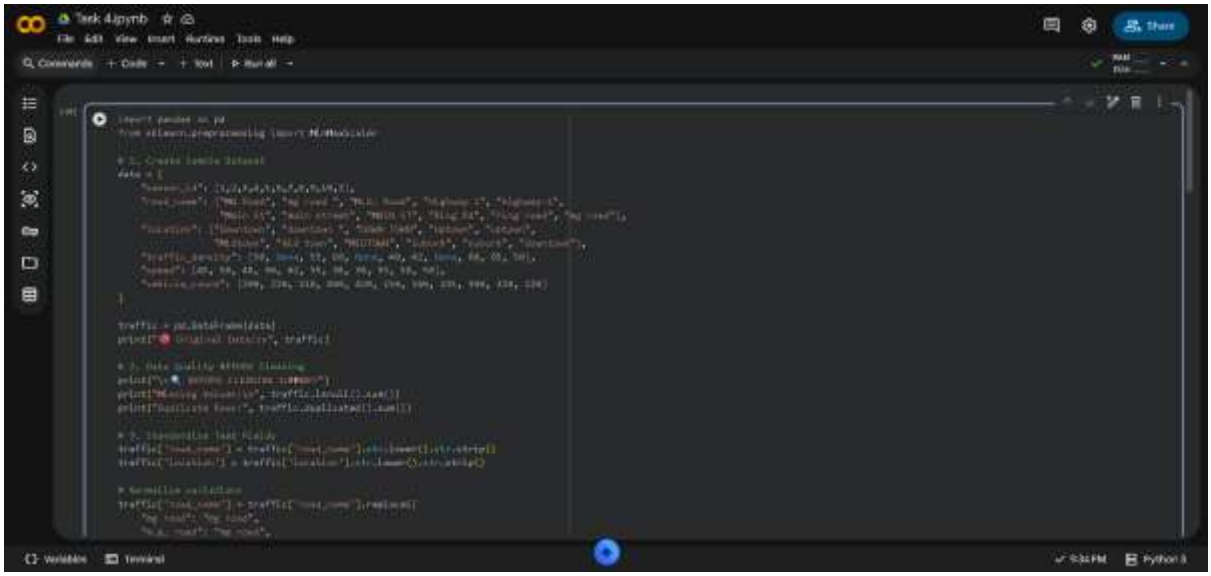
traffic = pd.read_csv("traffic_data.csv")

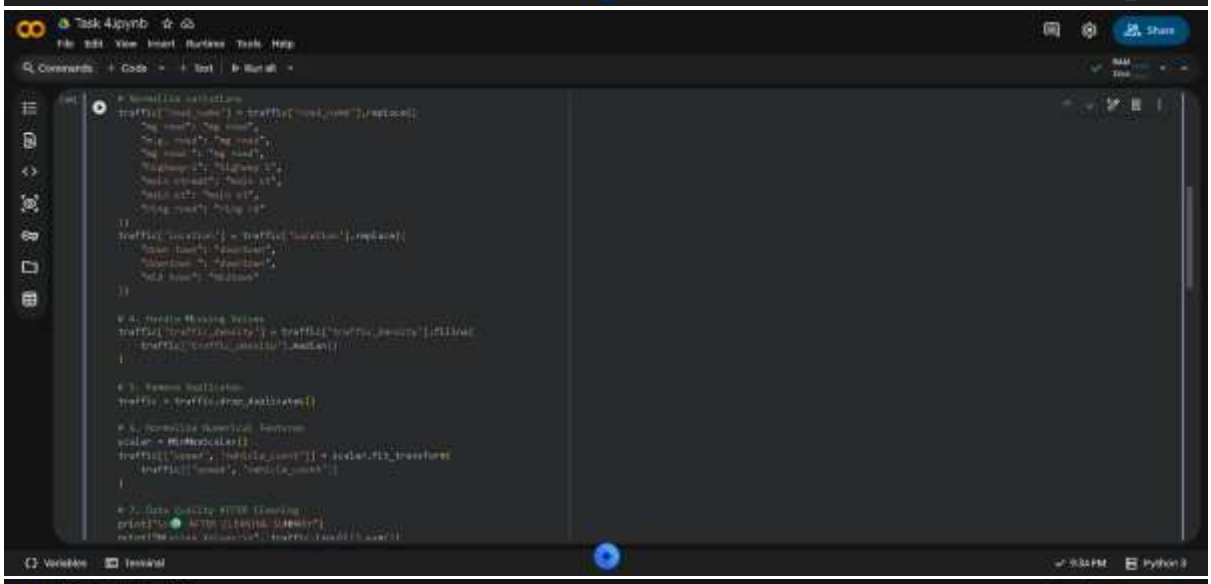
print(traffic.describe())

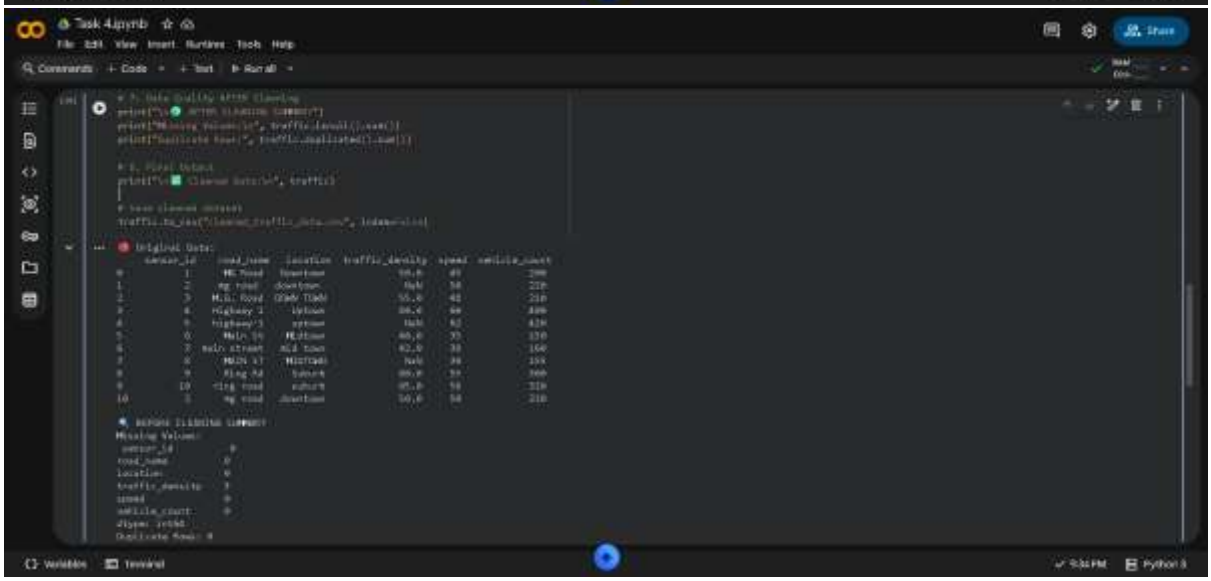
Expected Output:

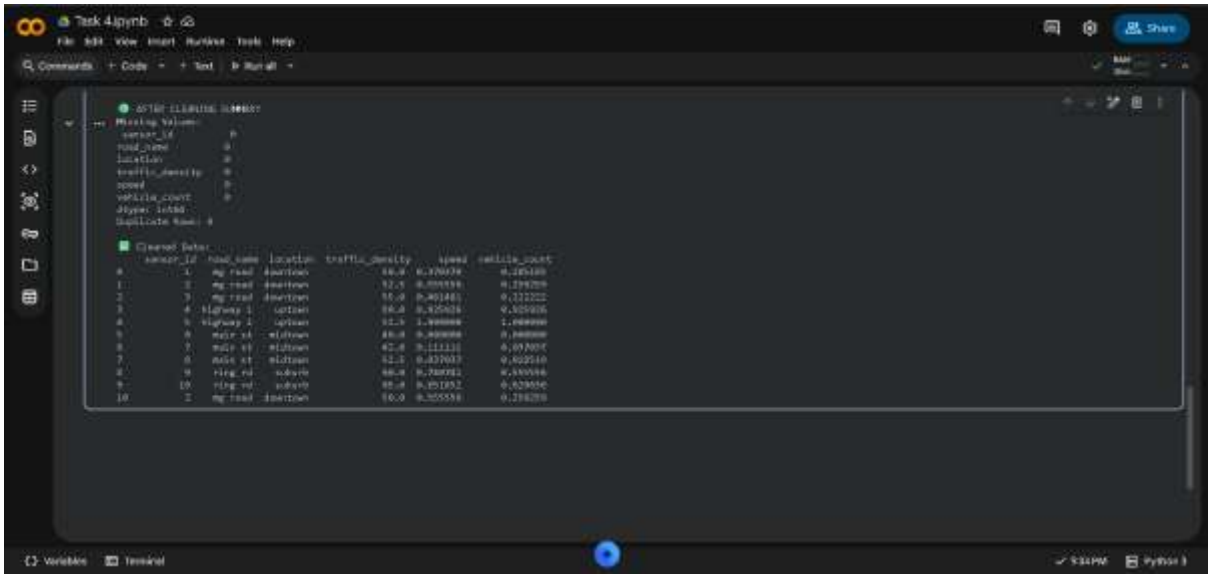
• A cleaned and normalized traffic dataset with an accompanying

data-quality improvement summary.









Task 5 – (Social Media Text Data Cleaning and Feature Extraction)

• Task:

Use AI-assisted Python scripting to clean and preprocess social media text data for sentiment analysis.

Instructions:

• Remove duplicate posts and null text entries.

• Clean text by removing URLs, special characters, and extra spaces.

• Convert text to lowercase and perform basic tokenization.

• Remove stopwords and perform stemming or lemmatization. Generate basic text features such as word count and sentiment score.

Sample Code:

```
import pandas as pd

import re

posts = pd.read_csv("social_media_posts.csv")

print(posts.head())
```

Expected Output:

A cleaned text dataset with extracted features suitable for sentiment analysis models.

Task 5.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
!pip install nltk textblob

Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: textblob in /usr/local/lib/python3.12/dist-packages (0.19.0)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
```

```
import pandas as pd
import re
from sklearn.feature_extraction.text import ENGLISH_STOP_WORDS
from nltk.stem import PorterStemmer
from textblob import TextBlob

# 1. Create Sample Dataset
data = {
    "post_id": [1,2,3,4,5,6,7,8,9,10],
    "text": [
        "I love this product! 😍 http://example.com",
        "Worst service ever!!! 😡",
        "None",
        "Amazing experience, will buy again!!!",
        "Check this out: https://test.com #awesome",
        "Not good, very bad service",
        "Ok product, nothing special",
        "I LOVE THIS PRODUCT! 😍 http://example.com",
        "Worst service ever!!! 😡"
        ]
    }
```

Variables Terminal

Task 5.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
posts = pd.DataFrame(data)
print("📦 Original Data:\n", posts)

# 2. Remove Duplicates & Nulls
posts = posts.drop_duplicates(subset='text')
posts = posts.dropna(subset=['text'])

# 3. Clean Text
def clean_text(text):
    text = text.lower()
    text = re.sub(r"https?\S+", "", text)       # remove URLs
    text = re.sub(r"[^a-z\s]", "", text)        # remove special chars
    text = re.sub(r"\s+", " ", text).strip()    # remove extra spaces
    return text

posts['clean_text'] = posts['text'].apply(clean_text)

# 4. Tokenization
posts['tokens'] = posts['clean_text'].apply(lambda x: x.split())

# 5. Remove Stopwords
posts['tokens'] = posts['tokens'].apply(
    lambda words: [w for w in words if w not in ENGLISH_STOP_WORDS]
)

# 6. Stemming
stemmer = PorterStemmer()
posts['tokens'] = posts['tokens'].apply(
    lambda words: [stemmer.stem(w) for w in words]
)

# Join tokens back
posts['processed_text'] = posts['tokens'].apply(lambda x: " ".join(x))
```

Variables Terminal

Task 5.ipynb

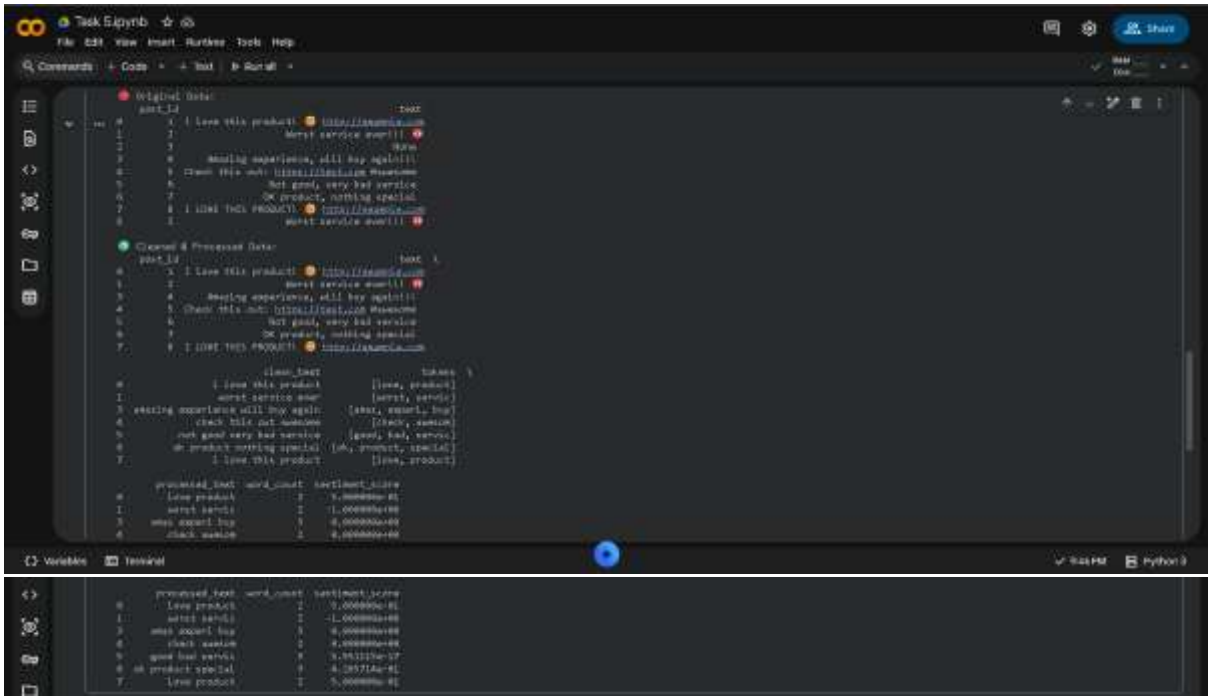
File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
# Join tokens back
posts['processed_text'] = posts['tokens'].apply(lambda x: " ".join(x))

# 7. Feature Extraction

# Word count
posts['word_count'] = posts['processed_text'].apply(lambda x: len(x.split()))

# Sentiment score
posts['sentiment_score'] = posts['processed_text'].apply(
    lambda x: TextBlob(x).sentiment.polarity
)

# 8. Final Output
print("\n✨ Cleaned & Processed Data:\n", posts)

# Save File
posts.to_csv("cleaned_social_media_posts.csv", index=False)
```

```
📦 Original Data:
   post_id                                                text
0        1  I love this product! 😍 http://example.com
1        2                          Worst service ever!!! 😡
2        3                                             None
3        4      Amazing experience, will buy again!!!
4        5  Check this out: https://test.com #awesome
5        6                     Not good, very bad service
6        7                        Ok product, nothing special
7        8  I LOVE THIS PRODUCT! 😍 http://example.com
8        9                          Worst service ever!!! 😡

✨ Cleaned & Processed Data:
   post_id                                                text  \
```

Variables Terminal


Task 5.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Original Data:
post_id text
0 1 I love this product! http://example.com
1 2 Worst service ever!!! 😡
2 3 Okay
3 4 Amazing experience, will buy again!!!
4 5 Check this out: https://test.com #awesome
5 6 Not good, very bad service
6 7 OK product, nothing special
7 8 I LOVE THIS PRODUCT! http://example.com
8 9 Worst service ever!!! 😡
Cleaned & Processed Data:
post_id text \
0 1 I love this product! http://example.com
1 2 Worst service ever!!! 😡
3 4 Amazing experience, will buy again!!!
4 5 Check this out: https://test.com #awesome
5 6 Not good, very bad service
6 7 OK product, nothing special
7 8 I LOVE THIS PRODUCT! http://example.com
clean_text tokens \
0 i love this product [love, product]
1 worst service ever [worst, servic]
3 amazing experience will buy again [amaz, experi, buy]
4 check this out awesome [check, awesom]
5 not good very bad service [good, bad, servic]
6 ok product nothing special [ok, product, special]
7 i love this product [love, product]
processed_text word_count sentiment_score
0 love product 2 5.000000e-01
1 worst servic 2 -1.000000e+00
3 amaz experi buy 3 0.000000e+00
4 check awesom 2 0.000000e+00
5 good bad servic 3 3.951111e-17
6 ok product special 3 4.285714e-01
7 love product 2 5.000000e-01
Variables Terminal
9:44 PM Python 3